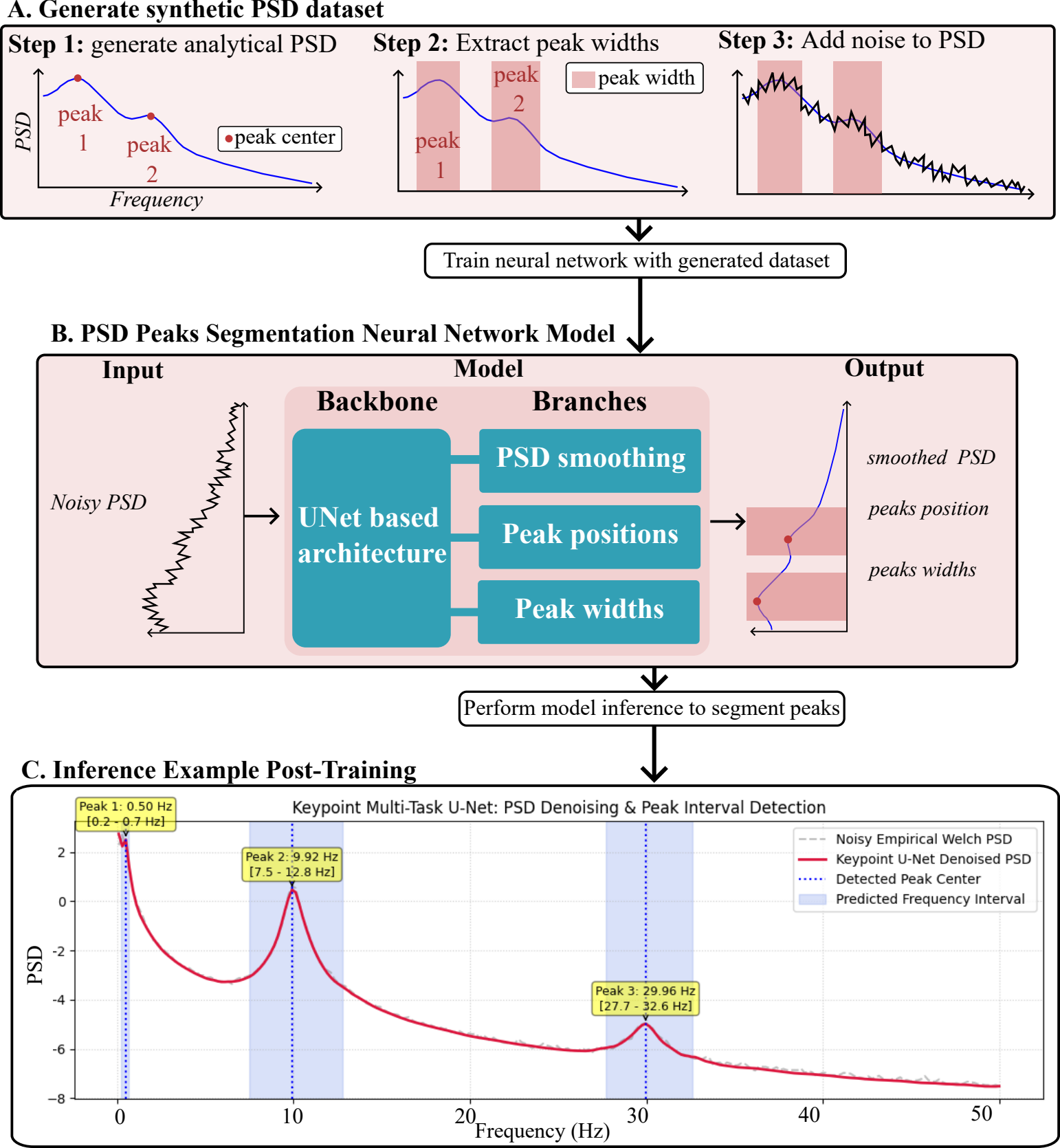

A. Generate synthetic PSD dataset
Step 1: generate analytical PSD
PSD
peak 1
peak 2
peak center
Frequency
Step 2: Extract peak widths
peak 1
peak 2
peak width
Step 3: Add noise to PSD
Train neural network with generated dataset
B. PSD Peaks Segmentation Neural Network Model
Input
Model
Output
Backbone
Branches
Noisy PSD
UNet based architecture
PSD smoothing
Peak positions
Peak widths
smoothed PSD
peaks position
peaks widths
Perform model inference to segment peaks
C. Inference Example Post-Training
Keypoint Multi-Task U-Net: PSD Denoising & Peak Interval Detection
Peak 1: 0.50 Hz [0.2 - 0.7 Hz]
Peak 2: 9.92 Hz [7.5 - 12.8 Hz]
Peak 3: 29.96 Hz [27.7 - 32.6 Hz]
Noisy Empirical Welch PSD
Keypoint U-Net Denoised PSD
Detected Peak Center
Predicted Frequency Interval
PSD
Frequency (Hz)